

Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics | OpenStax: OpenStax Viewer

COURSE AT A GLANCE

Review Content

REVIEW

Primary: [Intermediate Algebra 2e 3.2 Slope of a Line](#); [Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#); [Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation](#)

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation and write an equation from given information.
- I can apply a linear function to a context.
- I can identify the solution of a linear system as an intersection point.
- I can solve a simple two-variable linear system using an appropriate method.
- I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

Primary: [Intermediate Algebra 2e 3.5 Relations and Functions](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#)
Supplemental: [Intermediate Algebra 2e 3.6 Graphs of Functions](#); [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

UNIT LEARNING OBJECTIVES

- I can evaluate a function from a formula, graph, or table.
- I can determine whether a relation is a function.
- I can determine domain and range from a graph or table.
- I can solve for an input that produces a specified output.
- I can interpret inputs, outputs, domain, and range in context.
- I can calculate and interpret average rate of change.
- I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

Primary: [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#); [6.2 Factor Trinomials](#); [6.3 Factor Special Products](#); [6.5 Polynomial Equations](#)

UNIT LEARNING OBJECTIVES

- I can add and subtract polynomials.
- I can multiply polynomials.
- I can factor out a greatest common factor.
- I can factor a quadratic trinomial.
- I can factor a difference of squares and other required special cases.
- I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

Primary: [7.1 Multiply and Divide Rational Expressions](#); [7.2 Add and Subtract Rational Expressions](#); [7.4 Solve Rational Equations](#)

UNIT LEARNING OBJECTIVES

- I can state excluded values for a rational expression.
- I can simplify a rational expression while preserving its restrictions.
- I can multiply and divide rational expressions.
- I can add and subtract rational expressions using a common denominator.
- I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

Primary: [Intermediate Algebra 2e 8.3 Simplify Rational Expressions](#); [Intermediate Algebra 2e 8.2 Simplify Radical Expressions](#); [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)
Supplemental: [College Algebra 2e 2.6 Other Types of Equations](#)

UNIT LEARNING OBJECTIVES

- I can simplify expressions containing negative exponents.
- I can rewrite expressions between radical and rational-exponent forms.
- I can simplify radical expressions.
- I can apply exponent rules to expressions with integer and rational exponents.
- I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

Primary: [9.6 Graph Quadratic Functions Using Properties](#); [9.7 Graph Quadratic Functions Using Transformations](#); [6.5 Polynomial Equations](#); [9.1 Solve Quadratic Equations Using the Square Root Property](#); [9.2 Solve Quadratic Equations by Completing the Square](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#); [9.5 Solve Applications of Quadratic Equations](#)
Supplemental: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

UNIT LEARNING OBJECTIVES

- I can identify key features of a quadratic function from its graph or equation.
- I can connect standard, factored, and vertex forms to features of a parabola.
- I can solve a quadratic equation by factoring.
- I can solve a quadratic equation using the square-root property.
- I can solve a quadratic equation by completing the square.
- I can solve a quadratic equation using the quadratic formula.
- I can choose an efficient method for solving a quadratic equation.
- I can graph a quadratic function using its vertex, intercepts, and symmetry.
- I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

Primary: [8.8 Use the Complex Number System](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

UNIT LEARNING OBJECTIVES

- I can perform basic arithmetic with complex numbers.
- I can express the nonreal solutions of a quadratic equation using complex numbers.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary:** [Intermediate Algebra 2e 3.2 Slope of a Line](#); [Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#); [Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation](#)

M21-REV-001

I can determine and interpret the slope of a line.

OpenStax: **Primary:** [3.2 Slope of a Line](#)

SUPPORTING SKILLS

- Determine slope from a graph.
SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12
- Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-REV-002

I can graph a line from an equation and write an equation from given information.

OpenStax: **Primary:** [3.1 Graph Linear Equations in Two Variables](#)

SUPPORTING SKILLS

- Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M21-REV-003

I can apply a linear function to a context.

OpenStax: **Primary:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- I** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET
- R** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-REV-004

I can identify the solution of a linear system as an intersection point.

OpenStax: **Primary:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- R** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

OpenStax: **Primary:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- R** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- R** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- R** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- R** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

OpenStax: **Primary:** [5.2 Properties of Exponents and Scientific Notation](#)

SUPPORTING SKILLS

- R** Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- R** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- R** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12

Functions and Function Interpretation

REQUIRED

OpenStax coverage: **Primary:** [Intermediate Algebra 2e 3.5 Relations and Functions](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#)

Supplemental: [Intermediate Algebra 2e 3.6 Graphs of Functions](#); [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

M21-FUN-001

I can evaluate a function from a formula, graph, or table.

OpenStax: **Primary:** [3.5 Relations and Functions](#)

Supplemental: [3.6 Graphs of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · first introduced in middle school
 - D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
 - A** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ · first introduced in Math 12
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M21-FUN-002

I can determine whether a relation is a function.

OpenStax: **Primary:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
 - D** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12
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M21-FUN-003

I can determine domain and range from a graph or table.

OpenStax: **Primary:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
 - D** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12
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M21-FUN-004

I can solve for an input that produces a specified output.

OpenStax: **Primary:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT

M21-FUN-005

I can interpret inputs, outputs, domain, and range in context.

OpenStax: **Primary:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Restrict domain to values meaningful in a context.
SK-FUNC-CONTEXT-DOMAIN · first introduced in Math 12
- D** Interpret the range of a function in a context.
SK-FUNC-CONTEXT-RANGE · first introduced in Math 12
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12

M21-FUN-006

I can calculate and interpret average rate of change.

OpenStax: **Primary:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- I** Calculate average rate of change over an interval.
SK-FUNC-AROC
- I** Interpret average rate of change with contextual units.
SK-FUNC-AROC-INTERPRET
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-FUN-007

I can compare functions represented by formulas, graphs, or tables.

OpenStax: **Primary:** [Intermediate Algebra 2e 3.5 Relations and Functions](#)

Supplemental: [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

SUPPORTING SKILLS

- I** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP
- I** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH
- I** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH

Polynomials and Factoring

REQUIRED

OpenStax coverage: **Primary:** [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#); [6.2 Factor Trinomials](#); [6.3 Factor Special Products](#); [6.5 Polynomial Equations](#)

M21-POL-001

I can add and subtract polynomials.

OpenStax: **Primary:** [5.1 Add and Subtract Polynomials](#)

SUPPORTING SKILLS

- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
- D** Identify like polynomial terms by variable part and exponent.
SK-POLY-LIKE-TERMS · first introduced in Math 12
- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-002

I can multiply polynomials.

OpenStax: **Primary:** [5.3 Multiply Polynomials](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- D** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12
- D** Multiply a monomial by a polynomial.
SK-POLY-DISTRIBUTE · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-003

I can factor out a greatest common factor.

OpenStax: **Primary:** [6.1 Greatest Common Factor and Factor by Grouping](#)

SUPPORTING SKILLS

- R** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- D** Factor the greatest common monomial factor from a polynomial.
SK-FACTOR-GCF · first introduced in Math 12
- A** Determine the numerical and variable greatest common factor of polynomial terms.
SK-POLY-GCF · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-004

I can factor a quadratic trinomial.

OpenStax: **Primary:** [6.2 Factor Trinomials](#)

SUPPORTING SKILLS

- A** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- R** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- R** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-005

I can factor a difference of squares and other required special cases.

OpenStax: **Primary:** [6.3 Factor Special Products](#)

SUPPORTING SKILLS

- A** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- I** Factor a difference of two squares.
SK-FACTOR-DIFF-SQUARES
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-006

I can use factoring to determine the zeros of a polynomial.

OpenStax: **Primary:** [6.5 Polynomial Equations](#)

SUPPORTING SKILLS

- A** Factor a difference of two squares.
SK-FACTOR-DIFF-SQUARES · first introduced in Math 21
- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- I** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

Rational Expressions and Equations

REQUIRED

OpenStax coverage: **Primary:** [7.1 Multiply and Divide Rational Expressions](#); [7.2 Add and Subtract Rational Expressions](#); [7.4 Solve Rational Equations](#)

M21-RAT-001

I can state excluded values for a rational expression.

OpenStax: **Primary:** [7.1 Multiply and Divide Rational Expressions](#)

SUPPORTING SKILLS

- I** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS
- I** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

OpenStax: **Primary:** [7.1 Multiply and Divide Rational Expressions](#)

SUPPORTING SKILLS

- I** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS
- I** Factor numerators and denominators completely.
SK-RAT-FACTOR
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21
- A** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY · first introduced in Math 21

M21-RAT-003

I can multiply and divide rational expressions.

OpenStax: **Primary:** [7.1 Multiply and Divide Rational Expressions](#)

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- A** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

OpenStax: **Primary:** [7.2 Add and Subtract Rational Expressions](#)

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- I** Combine rational expressions over a common denominator.
SK-RAT-COMBINE
- I** Determine the least common denominator of rational expressions.
SK-RAT-LCD

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

OpenStax: **Primary:** [7.4 Solve Rational Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Clear denominators from a rational equation.
SK-RAT-CLEAR-DENOMINATORS
- I** Check candidate solutions against denominator restrictions.
SK-RAT-EXTRANEIOUS
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

Exponents and Radicals

REQUIRED

OpenStax coverage: **Primary:** [Intermediate Algebra 2e 8.3 Simplify Rational Exponents](#); [Intermediate Algebra 2e 8.2 Simplify Radical Expressions](#); [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)
Supplemental: [College Algebra 2e 2.6 Other Types of Equations](#)

M21-EXP-001

I can simplify expressions containing negative exponents.

OpenStax: **Primary:** [8.3 Simplify Rational Exponents](#)

SUPPORTING SKILLS

- I** Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE
- A** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- A** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12

M21-EXP-002

I can rewrite expressions between radical and rational-exponent forms.

OpenStax: **Primary:** [8.3 Simplify Rational Exponents](#)

SUPPORTING SKILLS

- I** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL
- I** Convert between radical and rational-exponent notation.
SK-RAD-CONVERT
- I** Read radical and rational-exponent notation and identify the index, radicand, base, numerator, and denominator.
SK-RAD-NOTATION-READ

M21-EXP-003

I can simplify radical expressions.

OpenStax: **Primary:** [8.2 Simplify Radical Expressions](#)

SUPPORTING SKILLS

- A** Express a whole number as a product of prime factors.
SK-NUM-PRIME-FACTOR · inherited from middle school
- I** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT

I can apply exponent rules to expressions with integer and rational exponents.

OpenStax: **Primary:** [8.3 Simplify Rational Exponents](#)

SUPPORTING SKILLS

- A** Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE · first introduced in Math 21
- D** Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- D** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- D** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21
- I** Apply the zero-exponent rule.
SK-EXP-ZERO

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

OpenStax: **Primary:** [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)

Supplemental: [College Algebra 2e 2.6 Other Types of Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT
- I** Check a candidate solution for extraneous results.
SK-RAD-CHECK
- I** Isolate a radical expression before solving.
SK-RAD-ISOLATE

Quadratic Functions and Equations

REQUIRED

OpenStax coverage: **Primary:** [9.6 Graph Quadratic Functions Using Properties](#); [9.7 Graph Quadratic Functions Using Transformations](#); [6.5 Polynomial Equations](#); [9.1 Solve Quadratic Equations Using the Square Root Property](#); [9.2 Solve Quadratic Equations by Completing the Square](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#); [9.5 Solve Applications of Quadratic Equations](#)
Supplemental: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

M21-QUAD-001

I can identify key features of a quadratic function from its graph or equation.

OpenStax: **Primary:** [9.6 Graph Quadratic Functions Using Properties](#)

SUPPORTING SKILLS

- I** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY
- I** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS
- I** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX
- I** Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
SK-QUAD-VOCABULARY

M21-QUAD-002

I can connect standard, factored, and vertex forms to features of a parabola.

OpenStax: **Primary:** [9.7 Graph Quadratic Functions Using Transformations](#)

Supplemental: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

SUPPORTING SKILLS

- I** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT
- D** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- A** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-003

I can solve a quadratic equation by factoring.

OpenStax: **Primary:** [6.5 Polynomial Equations](#)

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- I** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS

M21-QUAD-004

I can solve a quadratic equation using the square-root property.

OpenStax: **Primary:** [9.1 Solve Quadratic Equations Using the Square Root Property](#)

SUPPORTING SKILLS

- I** Apply the square-root property with both signs.
SK-QUAD-SQUARE-ROOT
- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-005

I can solve a quadratic equation by completing the square.

OpenStax: **Primary:** [9.2 Solve Quadratic Equations by Completing the Square](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE

M21-QUAD-006

I can solve a quadratic equation using the quadratic formula.

OpenStax: **Primary:** [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

SUPPORTING SKILLS

- I** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT
- I** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA
- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-007

I can choose an efficient method for solving a quadratic equation.

OpenStax: **Primary:** [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

SUPPORTING SKILLS

- A** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- A** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA · first introduced in Math 21
- I** Select an efficient quadratic-solving method from equation structure.
SK-QUAD-METHOD-SELECT
- A** Apply the square-root property with both signs.
SK-QUAD-SQUARE-ROOT · first introduced in Math 21

I can graph a quadratic function using its vertex, intercepts, and symmetry.OpenStax: **Primary:** [9.6 Graph Quadratic Functions Using Properties](#)**SUPPORTING SKILLS**

- A** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21
- D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

I can create and interpret a quadratic model for a projectile, area, or optimization problem.OpenStax: **Primary:** [9.5 Solve Applications of Quadratic Equations](#)**SUPPORTING SKILLS**

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- I** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary:** [8.8 Use the Complex Number System](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

M21-EXT-001

I can perform basic arithmetic with complex numbers.

OpenStax: **Primary:** [8.8 Use the Complex Number System](#)

SUPPORTING SKILLS

- I** Simplify integer powers of the imaginary unit.
SK-CPLX-I-POWERS
- I** Add, subtract, and multiply complex numbers.
SK-CPLX-OPERATE

M21-EXT-002

I can express the nonreal solutions of a quadratic equation using complex numbers.

OpenStax: **Primary:** [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

SUPPORTING SKILLS

- A** Simplify integer powers of the imaginary unit.
SK-CPLX-I-POWERS · first introduced in Math 21
- I** Express a quadratic solution with a negative discriminant in complex form.
SK-CPLX-QUAD-SOLUTION
- A** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT · first introduced in Math 21